Teaching Portfolio

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Teaching Identity and the Spark

Teaching is a profound passion that also defines my professional identity. My journey began with tutoring high school peers, evolving into roles as a part time Assistant Lecturer after my undergraduate degree in Nepal. The joy of facilitating understanding and receiving positive feedback fueled my commitment to education. This passion deepened as I observed students' progress in the courses I taught. For me, teaching is not just a profession but a calling where I thrive, dedicated to fostering student growth and success.

Guided by Lee Shulman's view that teaching involves transforming knowledge into forms that are pedagogically powerful and adaptive to students, teaching has become a professional commitment. I teach primarily undergraduate and graduate students in Industrial Engineering, Operations Research, analytics, and decision-making courses such as Engineering Economics, Optimization Methods with Applications, and Python for Data Science. Drawing on my experience as a part-time Assistant Lecturer in Nepal and as a Teaching Assistant at the University of South Florida, I use scaffolded explanations, guided reasoning, and real-world decision-making examples to help students engage deeply with quantitative methods rather than memorizing procedures. My research background in multi-objective optimization for sustainability applications, supported by agencies such as the U.S. Environmental Protection Agency and the U.S. Army Corps of Engineers, allows me to connect classroom learning with authentic societal challenges. This approach helps students develop confidence in analytical thinking, creativity in problem solving, and the ability to apply optimization and data-driven methods to complex engineering and policy decisions.

Teaching Philosophy

My teaching philosophy is shaped by Lee S. Shulman's view that understanding is demonstrated through teaching as the disciplined act of transforming one's knowledge into instruction that learners can genuinely comprehend (Shulman, 1986). I interpret this transformation work as simultaneously intellectual and humane: it requires a rich repertoire of representations such as examples, analogies, explanations, and demonstrations. It also requires sustained attention to the preconceptions and misconceptions students bring from diverse backgrounds and levels of prior preparation. Complementing Shulman, Jerome Bruner argues that any subject can be taught effectively "in some intellectually honest form" at any stage of development (Metsämuuronen J, 2018). This challenges me to design learning pathways that preserve disciplinary integrity while making complexity accessible through careful sequencing and scaffolding.

In practice, I aim to be an instructional scaffold who designs environments where students construct understanding by doing, testing ideas through problem solving, learning collaboratively, and refining their reasoning through feedback rather than passively receiving information. This commitment is evidence-informed: a large meta-analysis in undergraduate STEM finds that active learning improves performance and substantially reduces failure rates compared with traditional lecturing, so I prioritize structured participation and frequent opportunities for application and reflection (Scott Freeman, 2014). I also work deliberately to create an inclusive climate of belonging and respect because motivation and persistence are fostered when learners perceive the environment as a place where they belong and feel safe and valued.

Finally, I treat formative assessment as a core equity practice: I build regular, low-stakes feedback loops so students can see what they understand, what they do not yet understand, and how to close that gap consistently with research evidence that strengthening frequent feedback can yield substantial learning gains (Paul Black, 1998).

Teaching Narrative

In my teaching narrative, I make explicit how my course artifacts operationalize my philosophy of transforming disciplinary knowledge into learnable and intellectually honest representations. I play the role of a scaffold to help students with diverse prior preparation grow, by linking each artifact to a specific learning goal, breaking complex problems into learnable pieces, and providing them options for instructional methods.

I begin each course with low-stakes diagnostics such as an ungraded pre-quiz, a short survey, and a concept check, to surface prior knowledge, misconceptions, and constraints, because students' incoming knowledge strongly shapes how they interpret new content and because early diagnosis allows me to adjust pacing, target prerequisite gaps, and allocate support efficiently. I then align learning objectives, assessments, and in-class activities so that students repeatedly practice the exact forms of reasoning I expect them to demonstrate on problem sets, exams, and projects.

During class, I interleave concise explanations and worked examples with structured active-learning routines such as individual thinking & paired reasoning, and case discussions, because meta-analytic evidence in undergraduate STEM shows that active learning improves performance and reduces failure rates, and because high-quality active learning paired with inclusive practice can narrow achievement gaps (Scott Freeman, 2014).

To manage this effectively in both large lectures and smaller sections, I use real-time polling and peer discussion to make student thinking visible and to guide immediate instructional adjustments. Although research cautions that response systems must be used intentionally but also indicates they can improve learning and classroom climate when grounded in good questioning and interactive use (Caldwell, 2007). To bridge theory and practice, I use authentic, data-driven decision scenarios and scaffold students into a semester-long project sequence, from proposal to deliverables milestone to final model & analysis, often implemented via programming plus a written report & presentation. Drawing on evidence that problem-driven methods such exercise problems, projects, and case-based learning can support motivation and meaningful engagement when implementation is well designed. Across major assignments and projects, I increase transparency by explicitly communicating purpose, task, and evaluation criteria and by using clear rubrics, because transparent design has been shown to improve students' success-related outcomes, including confidence and belonging, with particularly strong benefits for underserved students.

Throughout the term, I employ frequent formative feedback loops such as low stakes checks, targeted comments on drafts, and opportunities to revise, because research reviews show that strengthening feedback practices can yield substantial learning gains, and because retrieval practice with feedback helps students both retain core ideas and direct their studying more effectively. Consistent with the goal of an equitable learning environment, I also establish shared discussion norms that value students' ideas and support respectful critique, so students feel safe to contribute and can learn through making their thinking public.

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Teaching Experience

Nepal and the United States

My official teaching journey began after my undergraduate degree, when I held a part-time assistant lecturer position in *Material Science & Metallurgy*, focusing on sustainable materials for engineering design. After that, I worked as a teaching assistant (TA) at the University of South Florida (USF), where I taught diverse courses like *Engineering Economic Analysis* for undergraduates, *Python for Data Science* for both undergraduates and master's students and *Optimization Methods with Applications* for graduate students. Each course presents unique content, and my teaching approach remained unified through various pillars mentioned in the philosophy above. I was particularly responsible for delivering lectures, developing exam questions, leading Friday recitation classes which include teaching one hour of solving numerical examples and case studies, grading exams and holding office hours to assist students on clearing their doubts and helping complete their course projects.

Courses Taught and Responsibilities

As an Assistant Lecturer for Material Science and Metallurgy, I developed lab tutorial syllabus in accordance with course's learning goals of Nepal Engineering Council and University Grant Commission's guidelines, an accreditation and funding agencies in Nepal. I tailored questions for exams after delivering lectures to students throughout the semester. The letter from the head of department at the Institute of Engineering, Tribhuvan University stating I have successfully completed my teaching duties is in the Appendix.

In Engineering Economics, I focused on cultivating decision-making through real-world data and showed relevance of the course in their personal finance. I guide students to view problems not just through equations, but through the lens of logical reasoning and long-term impact in their financial planning. I set clear formatting standards and enforce attention to detail in assignments to reflect the level of discipline and professionalism required in engineering practice. I also incorporate self and peer evaluation to help students develop critical reflection skills, an essential part of continuous professional growth. Moreover, I have worked on developing exam questions to help professors on this course.

In Optimization Methods with Applications, I encouraged students to go beyond textbook methods. While grading, I recognize and reward alternative, logically sound solutions that reflect creative approaches to mathematical modeling. I see great value in students making thoughtful assumptions, attempting novel formulations, and learning from missteps. Optimization is about accuracy, but also about exploration and insight, and I try to instill this idea into students during their journey of completing course project works.

In Python for Data Science, I actively involved students by coding in real time. Instead of reading through slides, I walk them through the logic of each solution interactively, showing the process behind debugging and refining code. This not only makes learning more dynamic and visual but also helps students develop the problem-solving mindset critically in data science. I emphasize that learning to program is not about memorizing syntax but about learning how to think systematically. Find the syllabus of all above courses in the Appendix.

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Summary Table of Courses Taught

S. N.	Course	Institution	Level	No. of Semesters	Role	Responsibilities
1	Engineering Economics	University of South Florida	Bachelor's	Total Seven Semesters	Teaching Assistant	Recitation lecture and hands-on practice, Grading, Office-hours
2	Optimization Methods with Applications	University of South Florida	Master's	Spring 2022	Teaching Assistant	Project advising and guidance, Grading, Office-hours
3	Python for Data Science	University of South Florida	Master's	Summer 2023	Teaching Assistant	Recitation and hands-on practice, Project advising and guidance, Grading, Office-hours
4	Material Science and Metallurgy	Tribhuvan University, Institute of Engineering	Bachelor's	Fall 2018 Spring 2019	Assistant Lecturer	Primary Instructor, Syllabus update, Exam question development, Lab experiments, Grading

Implemented Practical Pedagogy

Moreover, while fulfilling these responsibilities I have used active learning, flipped learning techniques, learning through audio visuals, and case-based & project-based learning. This experience honed my ability to teach effectively, and students also became curious about learning the course materials in depth. Then, I have provided constructive feedback using canvas and managed weekly recurring student interactions using both MS Teams and face-to-face settings and fostered quick responsiveness. A major milestone in my teaching experience was navigating the COVID-19 pandemic, which significantly changed the teaching landscape. Serving as a teaching assistant in both in-person and online settings during and after the pandemic taught me to leverage technology to create student-centered learning environments. Find the link to the video recording and syllabus of a hybrid class where I taught certain chapters of a course Engineering Economic Analysis in the Appendix. My teaching experience in both the United States and Nepal strengthened my ability to create inclusive learning environments across cultures, and working with diverse class sizes has equipped me to adapt my methods effectively.

A major focus in all my TA roles has also been on building collaborative and professional classroom environments. I've used tools to track student engagement, identify those needing encouragement, and support inclusive teamwork. I've handled challenging group dynamics by reinforcing good communication habits, such as involving all team members in emails and presentations, and promoting accountability within student teams. These soft skills are often overlooked, but they're foundational to success in real-world engineering roles. Ultimately, these experiences reinforced my belief that teaching is not about giving the right answers, it's about asking the right questions and empowering students to find their own solutions. I aim to mentor students into becoming confident, curious, and capable problem-solvers. By promoting creativity, professionalism, and collaboration, I strive to shape classrooms where students don't just learn, they grow into the engineers and thinkers the world needs.

Student Assessment, Teaching Evaluation and Self Reflection

Assessment Methods

I assess student learning through a variety of formative and summative methods, from low-stakes minute papers or in-class discussion prompts to formal assignments, exams, and projects. They are chosen so that different students have multiple ways to demonstrate their understanding. I normally design assessments that align directly with intended learning outcomes, ensuring that what I teach and how I assess are coherently linked to the course objectives. To serve diverse learners and foster an inclusive classroom, I offer alternative formats and flexible participation modes to accommodate different learning styles and backgrounds. I regularly review assessment outcomes like grades, common errors, and student comments to reflect on what worked and what didn't, and then adapt my teaching, especially by revising assignments, adding scaffolding or offering more support which improves student success. Through these practices, I strive to create a learning environment in which all students not only can succeed but clearly recognize their progress and build confidence in their abilities.

I also assess myself through mid-term surveys and end-of-semester evaluations which guide my continuous improvement, with students consistently noting my caring, accessible, and respectful approach.

Students Feedback and Comments

My TA supervisor during September 2025, Dr. Chilton, helped create a Canvas-based feedback survey to assess my teaching effectiveness. Thirty-one students responded, providing qualitative feedback. Here is the link to all thirty-one responses from students and survey questionnaires with a summary of responses that can be downloaded and viewed. This feedback highlighted my ability to break down complex concepts into digestible parts, create an inclusive and supportive classroom environment, and communicate clearly. Here are some selected writings from students:

- "Ashim's performance as a teaching assistant was better than that of professors, I had in my first degree... He has a strong ability to break things down into their smallest pieces and go bit-by-bit". - *Ryan Altizer, Electrical Engineering Undergraduate Student*
- "He made the recitation feel like a safe space to ask questions." - *Nguyen Nhu Quynh Van, Civil & Environmental Engineering Undergraduate Student*
- "He made these ideas easier to understand by breaking them into smaller parts and using examples from real life." - *Maxwell Ng, Civil & Environmental Engineering Undergraduate Student*

This feedback reinforced my belief that effective teaching requires scaffolding, breaking complex ideas into learnable parts, clarity, patience, and a genuine commitment to student success. The table below is the consolidation of qualitative responses from students (S) selected for every qualitative feature denoted by question number (Q).

Feature	Sampled anonymized quote (1 per feature)
Teaching clarity & explanation	S02 (Q9): "Ashim's explanations have really helped me … and I feel substantially more confident in planning…"
Problem-solving & worked examples	S14 (Q2): "He solved problems clearly and methodically, walking through the reasoning behind every step… This helped me understand not just how … but also why."
Engagement & delivery	S01 (Q3): "Ashim did a great job filling in and he kept the class engaged."

Organization & punctuality	S09 (Q4): "They were always punctual and organized … the recitation sessions always started and wrapped up on time…"
Accessibility & support	S01 (Q5): "He was very approachable and supportive."
Responsiveness & communication	S02 (Q4): "Generally, if reached out to, he was very quick to respond… and … get an answer … in a timely manner."
Approachability & inclusivity	S01 (Q8): "I felt very comfortable approaching Ashim. He was respectful, encouraging, and created a welcoming environment…"
Enthusiasm & positive attitude	S01 (Q7): "Yes, TA consistently showed enthusiasm and willingness to help students succeed."
Practical application & real-world relevance	S25 (Q9): "Ashim's instruction helped bridge theory with real-world applications… apply engineering economics to personal finance and … decision-making."
Knowledge & expertise	S03 (Q10): "He shows his knowledge very well … and makes sure that he is understood before he continues…"
Professionalism, respect & patience	S09 (Q5): "TA was incredibly professional … addressed my confusion and … uploaded corrections the same day…"
Pacing & time allocation	S25 (Q3): "His delivery of the lecture material was well-paced and interactive."
Instructional modality & logistics	S07 (Q10): "I believe he should always teach in person… it gets hard to understand … over bad Wi-Fi…"
Explicit suggestions for improvement	S04 (Q10): "Some more practice of the questions might have helped to better prepare him for recitation sessions…"

Summary Statistics

Engineering Economic Analysis EGN 3615 Summer 2025 and the Syllabus.

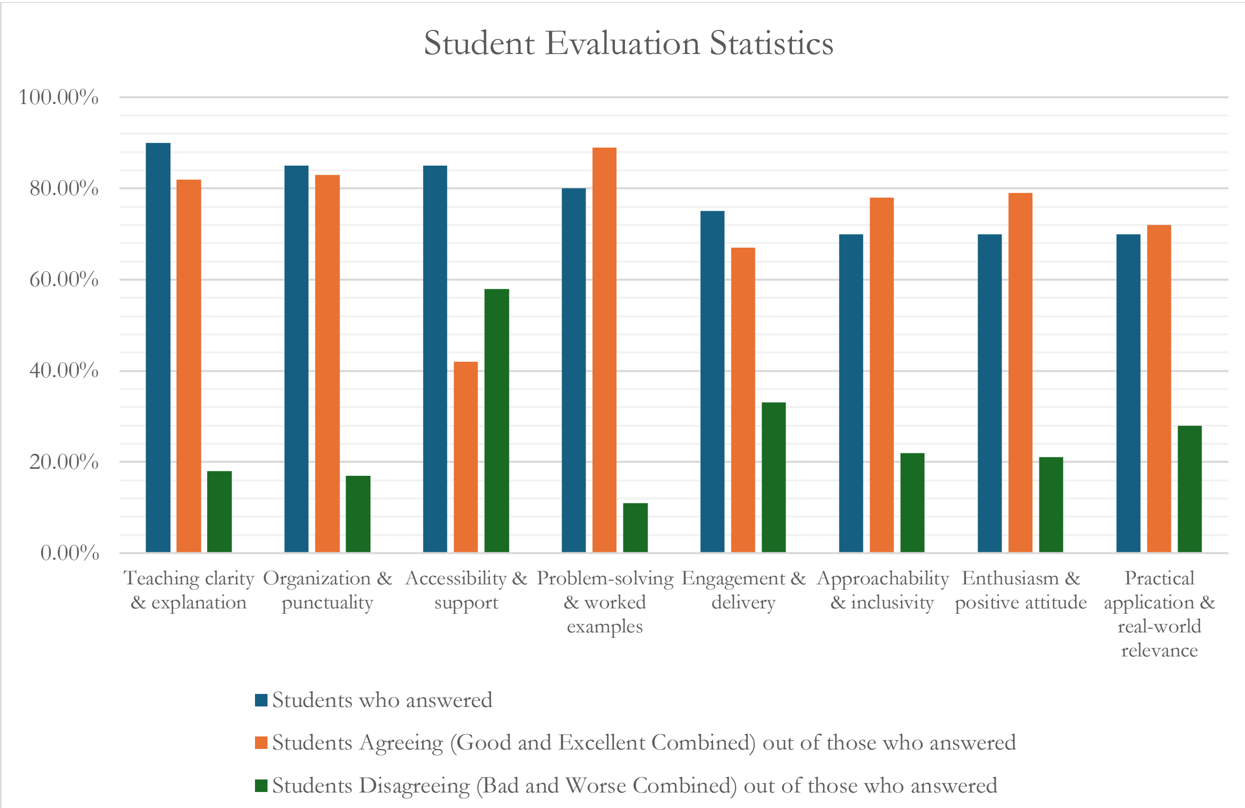
Average Student Score	88.23 %
Minimum Score	65.76 %
Maximum Score	100 %
Median Score	88.82 %

Find the statistics of the top 8 feature towards which students responded.

S.N.	Feature	Students who answered	Students Agreeing (Good and Excellent Combined) out of those who answered	Students Disagreeing (Bad and Worse Combined) out of those who answered
1	Teaching clarity & explanation	90%	82%	18%
2	Organization & punctuality	85%	83%	17%
3	Accessibility & support	85%	42%	58%
4	Problem-solving & worked examples	80%	89%	11%
5	Engagement & delivery	75%	67%	33%
6	Approachability & inclusivity	70%	78%	22%
7	Enthusiasm & positive attitude	70%	79%	21%

8	Practical application & real-world relevance	70%	72%	28%

Find the bar graph representation of this statistics below.



Applied Industry Experience and How it Shaped my Teaching

My professional experience in industry is a key asset that I integrate into my teaching. Having worked in supply chain and operations roles for multinational companies like Unilever and Coca-Cola, I bring firsthand industry insights into the classroom. This background allows me to enrich lectures and discussions with practical examples of how operations and analytics concepts drive value in real business settings. For instance, when teaching supply chain management, I draw on my experience in production operations to discuss challenges like demand forecasting, inventory optimization, and lean management. In analytics and decision science courses, I can cite instances where data-driven strategies improved efficiency and informed strategic decisions. By bridging theory with current practice, I help students appreciate the relevance of academic concepts and prepare them to apply their knowledge as future industry and business leaders.

My industry background also sharpens my approach to project-based learning. I often design assignments that mimic real-world engineering and business problems, such as optimizing a distribution network or analyzing sales data for insights, so that students gain experience solving practical challenges. This applied focus resonates strongly with students and underscores the value creation aspect of operations research and advanced analytics. My dual perspective as an educator and former industry practitioner enables me to mentor students on both the technical and managerial facets of decision-making, aligning with the mission to produce well-rounded, industry-ready graduates.

Teaching Innovation and Success

Innovation in teaching, for me, is not only the adoption of new tools for their own sake but it is the disciplined work of redesigning learning so that complex ideas become learnable, intellectually honest, and responsive to students' diverse preparation. In the Industrial Engineering courses I support, including Engineering Economics, Optimization Methods with Applications, and Python for Data Science, I focus on innovations that reduce barriers to participation, make student thinking visible, and accelerate feedback cycles so students can improve with guidance rather than guesswork. This approach is grounded in evidence that active learning in undergraduate STEM classes improves performance and reduces failure rates, with particularly strong equity implications when implementation is high quality (Scott Freeman, 2014).

A key access-oriented innovation I introduced is a low-friction communication and support workflow that allows students to request help and surface confusion without social risk. I use a canvas and online sign-up google docs form that streamlines student communication, supports more organized follow-up, and can preserve anonymity when students need it. This design choice is motivated by teaching-practice guidance showing that anonymous feedback channels can strengthen student voice and iteration during a term. Research suggests that anonymity can reduce fear of judgment and increase comfort engagement, especially for students who might otherwise stay silent. I also embed formative "muddiest point" style prompts to identify where learning is breaking down so review time can be targeted to the most consequential misconceptions (University, 2022).

In class and recitation, I aim to make expert reasoning visible rather than presenting only polished solutions. I use real-time problem solving on a tablet and live coding where appropriate to show how models are built iteratively, how assumptions are tested, and how debugging and revision are normal parts of quantitative work. Research on participatory live coding indicates that students value seeing the programming process unfold, including debugging strategies and good practices, and learning-science guidance on worked examples supports modeling solution steps to reduce cognitive load early in skill acquisition. In Python focused settings, I extend this with interactive notebook-based activities that connect explanation, code, and output in one environment, which has been associated with improved engagement and support for conceptual understanding when instruction is redesigned around hands-on notebook work.

To deepen learning and engagement, I complement explanation with structured active-learning cycles: brief concept questions, pair-trio reasoning, and polling to monitor understanding in real time. Evidence from long-running implementations of peer instruction shows gains in conceptual mastery and quantitative problem solving, and research reviews on response systems emphasize that the learning value comes from using questions to surface thinking and then responding instructionally. I also incorporate frequent low-stakes retrieval opportunities such as short quizzes, warm-ups, and cumulative checks, because testing and practice testing have strong evidence for improving long-term retention and helping students calibrate what they do and do not yet know.

Finally, I treat relevance and transparency as core innovations in engineering learning. I use authentic, data-driven decision scenarios and scaffold projects so students practice translating theory into models, analyses, and defensible recommendations with outcomes strongly aligned with the professional aims of industrial engineering. Reviews of project-based learning in higher education report both cognitive and affective benefits when projects are intentionally designed, and engineering-focused work on case-based learning highlights gains in engagement and skills such as critical thinking and problem solving. To ensure these experiences are equitable and navigable, I use transparent assignment design and rubrics including purpose, task, and criteria, because multi-institutional evidence shows that transparency strengthens academic confidence, belonging, and persistence, especially for underserved students. Together, these innovations are designed to produce not only stronger academic performance but also more confident, collaborative, and professionally prepared engineers (Pengyue Guo, 2020).

Future Teaching and Goals

Overall, I view teaching as a lifelong journey of growth and learning. By combining personalized, inclusive, state-of-the art pedagogy and technology-enhanced education with real-world applications and research-driven content, I aim to empower students to become critical thinkers, problem-solvers, and leaders. Additionally, my industry experience with multinational companies such as Unilever and Coca-Cola in supply chain and production operations management allows me to bring a professional and practical perspective to course development and instruction. Moreover, I plan to continuously learn and implement the recent developments in innovative teaching methods published in educational research journals such as *INFORMS Transactions on Education, Educational Researcher* and *International Journal of Educational Research,* and later contribute towards publishing in these transactions and journals.

As an aspiring assistant professor, I am eager to teach *Multi-objective Optimization under Uncertainty, Sustainable Supply Chain and Circular Economy, Predictive & Prescriptive Analytics, Decision Making with Deep Reinforcement Learning, Production Warehousing and Distribution Operations Management, Advanced Data Analytics and visualization with Python, Smart Systems for Real Time and Sustainable Decision Making etc.* informed by my research, teaching, projects and courses background.

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Appendix

1. Here is the link to the letter from the head of department at the Institute of Engineering, Tribhuvan University stating I have successfully completed my teaching duties.
2. Here is the MS TEAMS recording and drive link to video folder of a hybrid class where I taught certain chapters of a course Engineering Economic Analysis, along with its syllabus.
3. Thirty-one students responded, providing qualitative feedback. Here is the link to all thirty-one responses from students and survey questionnaires with a summary of responses that can be downloaded and viewed
4. Reference: Dr Jamie Chilton Email
5. Syllabus Engineering Economic Analysis
6. Syllabus Optimization Methods and Applications
7. Syllabus Python for Data Science